Harris Kaplan, Ph.D.

Department of Biology, University of Virginia, Charlottesville, VA
E-mail: harris.kap@gmail.com | Cell: 973-634-3340 | Web: bio.as.virginia.edu/people/harris-kaplan

RESEARCH INTERESTS

My lab investigates how neuronal circuits function in early life, to coordinate social and sleep behaviors of infants. I am trained as both a systems neuroscientist and a developmental molecular biologist.

RESEARCH EXPERIENCE & EDUCATION

2026- **Assistant Professor**, Department of Biology, University of Virginia, Charlottesville, VA

2020 **Postdoctoral Fellow in Neurobiology**, Department of Molecular & Cellular Biology, Harvard University, Cambridge, MA
Faculty Advisor: Catherine Dulac, Samuel W. Morris University Professor
Project: Developmental trajectories of hypothalamic cell types governing social behavior
Funding: Howard Hughes Medical Institute Fellow of the Jane Coffin Childs Fund

2019 **Ph.D. in Neuroscience**, Research Institute of Molecular Pathology (IMP), Vienna, Austria
Faculty Advisor: Manuel Zimmer, Group Leader
Project: Brain-wide neuronal population dynamics governing behavior in *C. elegans*

2012 **B.A. in Biology**, *summa cum laude*, New York University, New York City, NY
Faculty Advisor for Honors Thesis: Justin Blau, Professor of Biology and Neural Science
Project: Neuronal signaling mechanisms underlying circadian clock synchrony

PUBLICATIONS

Selected publications

1. **Kaplan HS**, Logeman BL, Zhang K, Yawitz TA, Santiago C, Sohail N, Talay M, Seo C, Naumenko S, Ho Sui SJ, Ginty DD, Ren B, Dulac C (2025). Sensory input, sex, and function shape hypothalamic cell type development. ***Nature*** 647, 157–168. DOI: 10.1038/s41586-025-08603-0

2. **Kaplan HS***, Salazar Thula O*, Khoss N, Zimmer M (2020). Nested neuronal dynamics orchestrate a behavioral hierarchy across timescales. ***Neuron*** 5 February 2020. DOI: 10.1016/j.neuron.2019.10.037. *Equal contribution.

3. Kato S*, **Kaplan HS***, Schrödel T*, Skora S, Lindsay TH, Yemini E, Lockery S, Zimmer M (2015). Global brain dynamics embed the motor command sequence of *Caenorhabditis elegans*. ***Cell***, 163(3), 656–669. DOI: 10.1016/j.cell.2015.09.034. *Equal contribution.

Additional published research

4. **Kaplan HS***, Horvath T*, Rahman MM*, Dulac C (2025). The neurobiology of parenting and infant-evoked aggression. ***Physiological Reviews*** 2025 Jan 1;105(1):315-381. DOI: 10.1152/physrev.00036.2023. *Equal contribution.

5. Sullivan ZA, Wong BJ, Kapoor V, Vincze PK, **Kaplan HS**, Giraudet P, Watson BR, Misherghi A, Lourido S, Moffitt JR, Dulac C (2025). Brain-immune interactions generate pathogen-specific sickness states. ***bioRxiv*** 2025.12.06.692770; DOI: https://doi.org/10.64898/2025.12.06.692770.
6. Logeman BL, Horvath PM, Talay M, Kapoor K, **Kaplan HS**, Dulac C (2025). Cell Type-Specific Hormonal Signaling Configures Hypothalamic Circuits for Parenting. ***bioRxiv*** 2025.12.11.693766; DOI: https://doi.org/10.64898/2025.12.11.693766
7. Nowakowski TJ, Nano PR, Matho KS, Chen X, Corrigan EK, Ding W, Gao Y, Heffel M, Jayakumar J, **Kaplan HS**, Kronman FN, Kovner R, Mannens CCA, Song M, Steyert MR, Venkatesan S, Wallace JL, Wang L, Werner JM, Zhang D, Yuan G, Zuo G, Ament SA, Colantuoni C, Dulac C, Fan R, Gillis J, Kriegstein AR, Krienen FM, Kim Y, Linnarsson S, Mitra PP, Pollen AA, Sestan N, Tward DJ, van Velthoven CTJ, Yao Z, Bhaduri A, Zeng H (2025). The new frontier in understanding human and mammalian brain development. ***Nature*** 647, 51–59 (2025). DOI: 10.1038/s41586-025-09652-1
8. Nimpf S, **Kaplan HS**, Nordmann GC, Cushion T, Keays DA (2024). Long-term, high-resolution in vivo calcium imaging in pigeons. ***Cell Reports Methods*** 2024 Feb 13:100711. DOI: 10.1016/j.crmeth.2024.100711
9. **Kaplan HS**[†]**,** Zimmer M (2020). Brain-wide representations of ongoing behavior: a universal principle? ***Current Opinion in Neurobiology*** 64, 60-69. DOI: 10.1016/j.conb.2020.02.008. [†]Corresponding author.
10. **Kaplan HS**, Nichols ALA, Zimmer M (2018). Opinion piece. Sensorimotor integration in *Caenorhabditis elegans*: a reappraisal towards dynamic and distributed computations. ***Philosophical Transactions of the Royal Society B: Biological Sciences*** 2018 Sep 10; 373 (1758). DOI: http://doi.org/10.1098/rstb.2017.0371
11. **Kaplan HS**, Zimmer M (2018). Preview article. Sensorimotor integration for decision making: how the worm steers. ***Neuron*** 97(2), 258-260. DOI: 10.1016/j.neuron.2017.12.042
12. Hums I, Riedl J, Mende F, Kato S, **Kaplan HS**, Latham R, Sonntag M, Traunmüller T, Zimmer M (2016). Regulation of two motor patterns enables the gradual adjustment of locomotion strategy in *Caenorhabditis elegans.* ***eLife*** 2016;5:e14116. DOI: 10.7554/eLife.14116
13. Collins B, **Kaplan HS**, Cavey M, Lelito KR, Bahle AH, Zhu Z, Macara AM, Roman G, Shafer OT, Blau J (2014). Differentially timed extracellular signals synchronize pacemaker neuron clocks. ***PLoS Biology***, 12(9): e1001959. DOI: 10.1371/journal.pbio.1001959

INVITED TALKS

Mount Sinai Neuroscience Seminar, 2025
Weill Cornell Emerging Leaders in Neuroscience Series, 2025
Max Planck Institute for Biological Intelligence Emerging Scientist Series, 2025
Max Planck Florida Institute NeuroMEETS Postdoc Seminar Series, 2024
Gordon Research Conference on Neural Development, Lucca, Italy, 2024
Gordon Research Conference on Molecular and Cellular Neurobiology, Lucca, Italy, 2024
Neuro Data Talk, Institute of Science and Technology, Austria, 2024
Neuronal Circuits conference, Cold Spring Harbor, 2024
4D Nucleome Neuroscience Subgroup, 2023
Harvard MCB department retreat, 2023
Harvard Neurodevelopment Workshop, 2023
Brain Initiative Cell Census Network, Developmental Subgroup, 2022

Harvard MCB department seminar series, 2022
Janelia Junior Scientist Workshop, Biological Optical Microscopy, 2019
Whole-Brain Imaging Workshop, International C. elegans Meeting, UCLA, 2019
The International Congress of Neuroethology, Brisbane, Australia, 2018
Gordon Research Conference on Neuromodulation, Hong Kong, 2015
Featured Invited Speaker, C. elegans Neuroscience Meeting, Madison, Wisconsin, 2014
Stanford University C. elegans community, 2014

TEACHING & ADVISING EXPERIENCE

Teaching Assistant
Responsibilities included developing new class materials, leading class discussions, grading assignments, running office hours sessions, and meeting with students individually
Molecular Basis of Behavior (Spring 2022), Harvard College, 91 students, 20 in my section.
Distinction in Teaching Award, based on student evaluations

Advising Experience
1 Master's student at Harvard University – Master's thesis (2024)
1 Undergraduate student at Harvard University – Senior thesis (2021-23)
2 Master's student at IMP Vienna, Austria – Master's thesis (2014-16; 2017-18)
Additional undergraduate (e.g. summer) and rotation students (2014-2025)

FELLOWSHIPS & AWARDS

Fellowships
NIH K99/R00 – scored, under consideration for funding
Howard Hughes Medical Institute Jane Coffin Childs Fellowship, 2020

Academic Recognition
Ph.D. Award, most outstanding thesis, Vienna BioCenter, 2019
Ph.D. Award, most outstanding thesis in Austria, ÖGMBT Life Science, 2019

Teaching Awards
Distinction in Teaching Award, Harvard College (TA for Molecular basis of behavior), 2022

Presentation Awards
Best Poster Award, Hypothalamus Gordon Research Conference, Ventura, CA, 2022
Best Poster Award, Austrian Neuroscience Association meeting, Vienna, Austria, 2017
Mattias Lauwers Award, best student seminar, Vienna BioCenter, 2015
Best Poster Award, Vienna BioCenter Ph.D. Student retreat, 2014

PROFESSIONAL SERVICE

Outreach
Harvard Medical School's Health Professions Recruitment and Exposure Program (HPREP) - 1-on-1 mentoring of high school students from underserved backgrounds. 1 student mentored per year for 3 months. 1 year on DEI committee, working to get high school students into lab internships. 2020-23.
Cientifico Latino's Graduate Student Mentorship Initiative – one-on-one mentorship in graduate student applications of four applicants from minoritized backgrounds. 2021 and 2024.

Leadership
Organizing committee of the Vienna Biocenter PhD Symposium. Organized the program, invited and hosted speakers. 2013.